



Profound

Method Statement Sonic Integrity Testing (Pile Integrity Testing)

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1 Principles of Sonic Integrity Testing



Figure 1.1. Introducing hammer blows

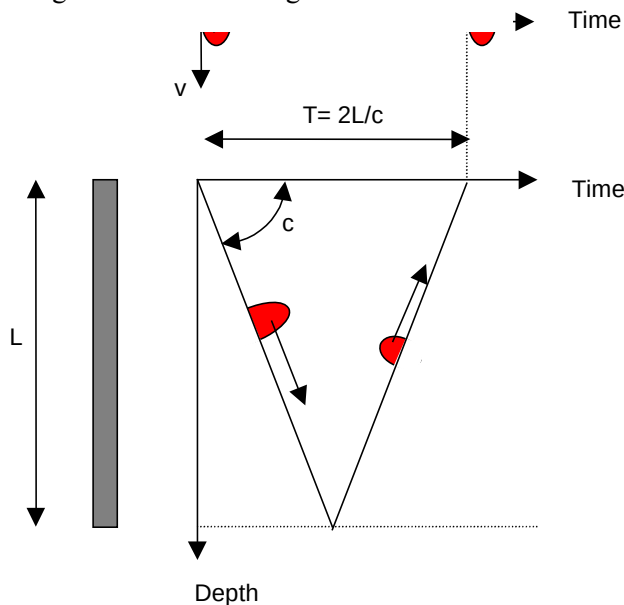


Figure 1.2

A stress wave (sound wave) is introduced into the pile by means of a hammer blow on the pile head (Fig.1.1).

This stress wave travels at the speed of sound (c) to the pile toe and reflects back to the pile head (Fig. 1.2).

The response of the pile head, as a result of the hammer blow and reflections, is measured with an acceleration transducer. The acceleration is integrated and presented as a velocity signal (v)

For each pile at least 3 hammer blows are applied to the pile head and the results are presented as 3 traces in a diagram (Fig. 3). To proof the quality of testing the 3 traces should be similar. The time (T) between the start of the hammer blow and the time of arrival of the reflection from the pile toe is measured. The pile length (L) is calculated with:

$$L = c.T/2$$

when the stress wave velocity (c) is known

To present the measuring results the time axis (t) is scaled to a length (depth) (L) axis with $L = c.t/2$. Due to shaft friction the toe reflection might be of small magnitude. To make the reflection visible, the measured signal is amplified.

To remove noise from the signals a filter value can be applied.

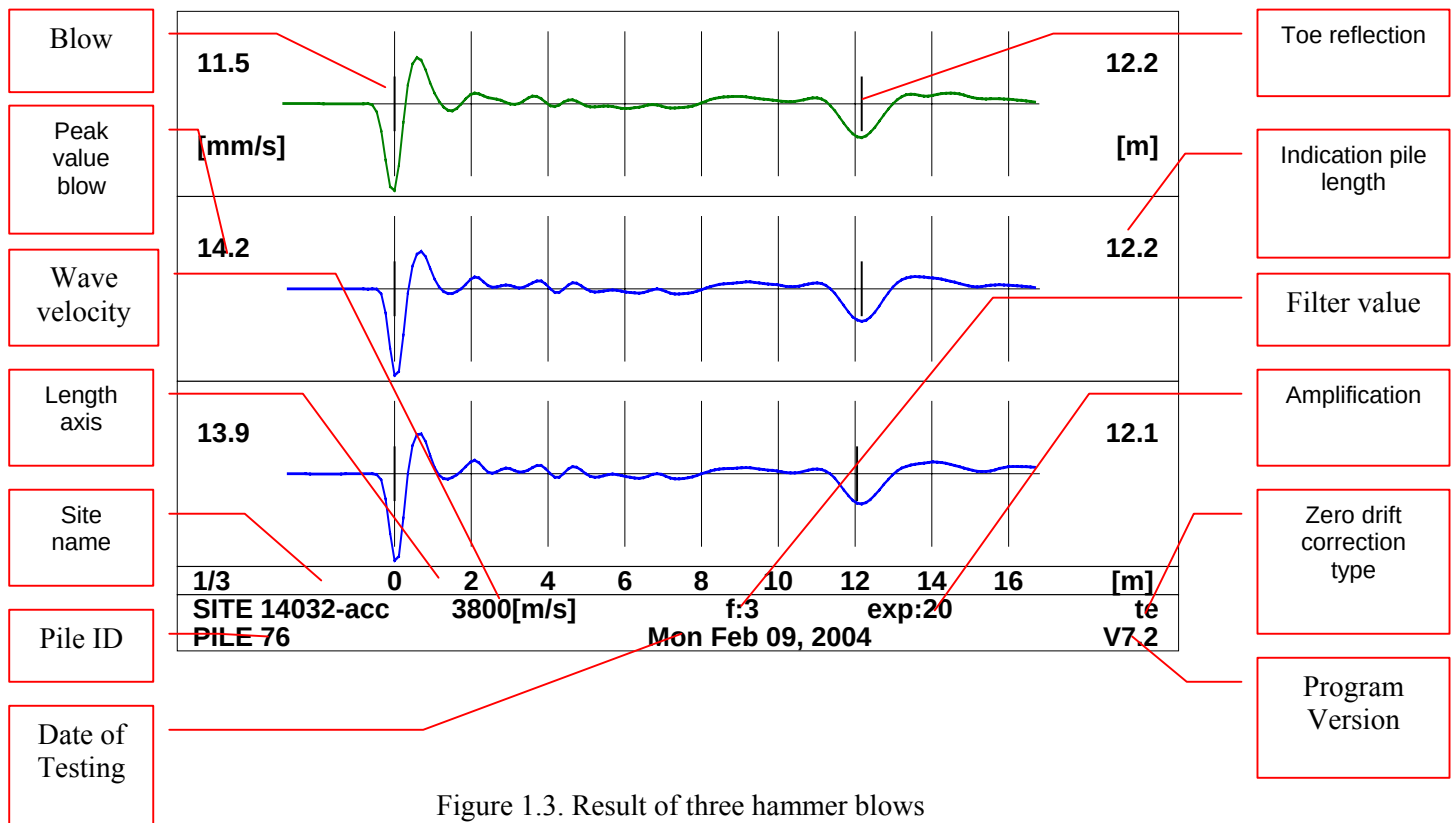


Figure 1.3. Result of three hammer blows

Pile integrity testing analysis is based on the one dimensional stress wave theory. Reflections generated by impedance changes (discontinuities) travel to the pile top and are recorded and analyzed

The impedance Z is defined as

$$Z = A \sqrt{E \cdot \rho}$$

In which

A = cross sectional area
 E = modulus of elasticity
 ρ = density

Any change in A , E , or ρ or a combination of them will generate a reflection from an impedance change (discontinuity). Potential causes for reflections are:

- ☐ Pile toe
- ☐ Dimensional changes
- ☐ Soil inclusions
- ☐ Cracks
- ☐ Joints
- ☐ Variations in concrete quality
- ☐ Variations in soil layers
- ☐ Overlap of reinforcement (heavily reinforced piles).



Limitations of sonic integrity testing are:

- ☐ Minor impedance changes are not detected
- ☐ Gradually increasing and decreasing pile diameters can not be detected
- ☐ Curved pile shapes can not be detected
- ☐ Small soil inclusions are not detected
- ☐ Local loss of reinforcement cover can not be detected
- ☐ Thickness of debris layer at pile toe can not be detected

2 Discontinuities in the pile shaft.

Reflections will occur when discontinuities in the pile cross section or pile material properties are present (Fig.2.1) From the time of occurrence of a reflection the location of the discontinuity can be determined. The sign of the reflection indicates an increase or decrease in the pile cross section or an increase or decrease in pile material quality.

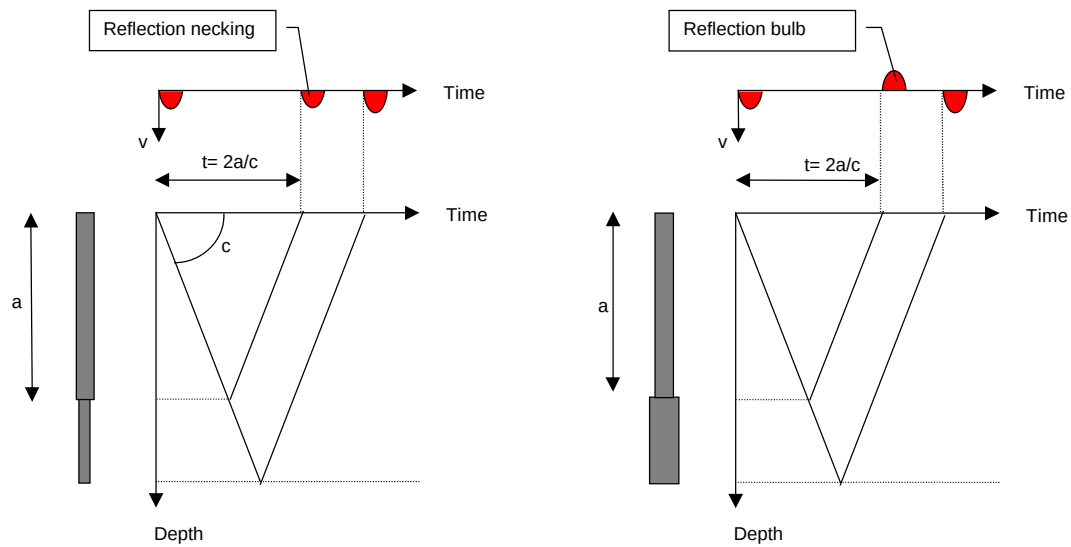


Figure 2.1.

3 Signal processing

3.1 Signal amplification

Shaft friction (see Appendix) has a strong influence on the pile integrity testing results and has to be taken into account. The signals measured on the pile top are amplified linearly or exponentially to overcome the reduction of the amplitude of the stress wave while traveling to the pile toe and back.

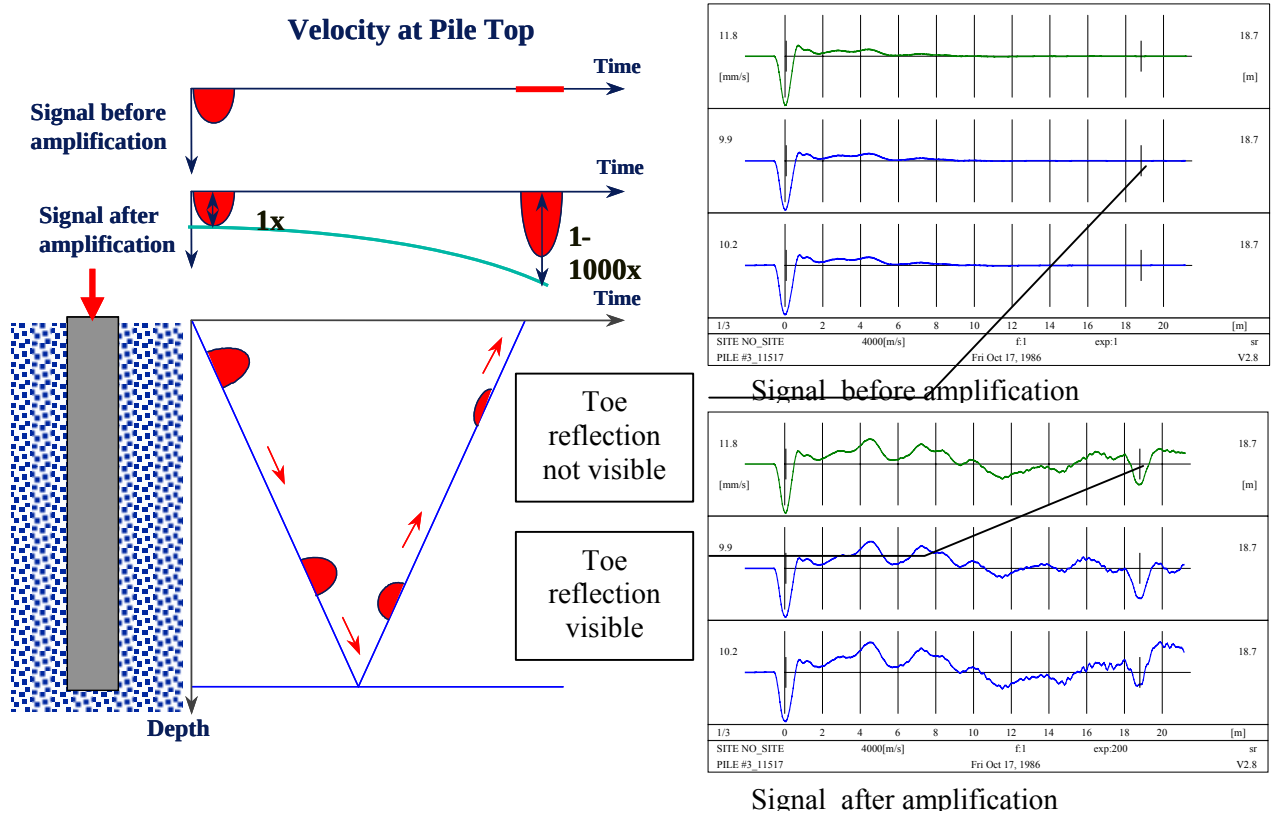


Figure 3.1, Signal amplification



3.2 Signal smoothing

Discontinuities near the pile top or extending reinforcement will generate high frequency reflections, which mask the global shape of the measured signals. To view the global shape of the signals smoothing is applied. It should be taken into account that by smoothing information about discontinuities is lost. For the final interpretation and presentation of the signals, smoothing should be reduced to a minimum .

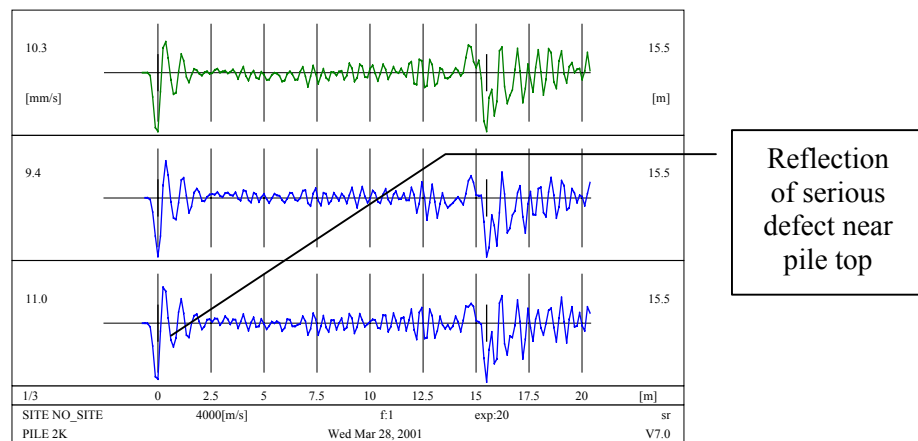


Figure 3.2, Unsmoothed signal

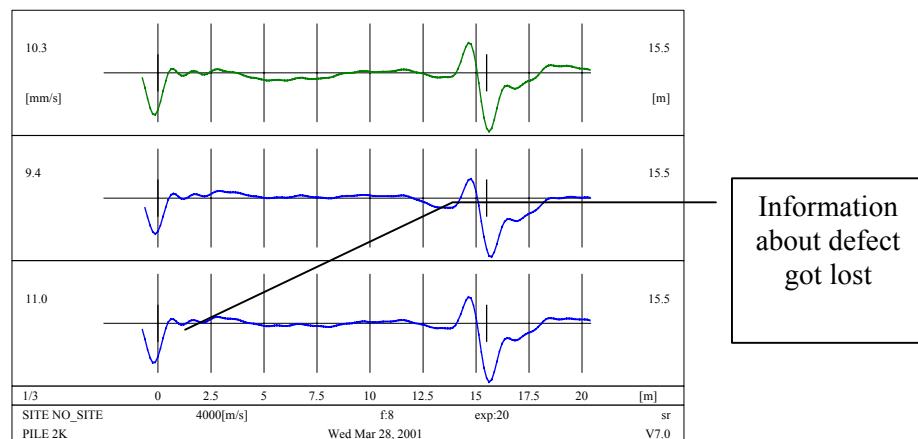


Figure 3.3, Smoothed signal



4 The Characteristic Signal

To differentiate between a change in soil resistance and a pile discontinuity, the **SIT** signal from a test pile is compared to a "characteristic signal" deemed to be representative of similar piles in similar soil conditions on site. (The **characteristic signal** can either be an **average** of a number of piles on site or the **SIT** signal of a reference pile chosen prior to testing.) If the test signal is different than the characteristic signal, then any impedance changes are due to the changing pile impedance and not characteristic of the site. Changes not found in the characteristic signal require further analysis to determine the cause.

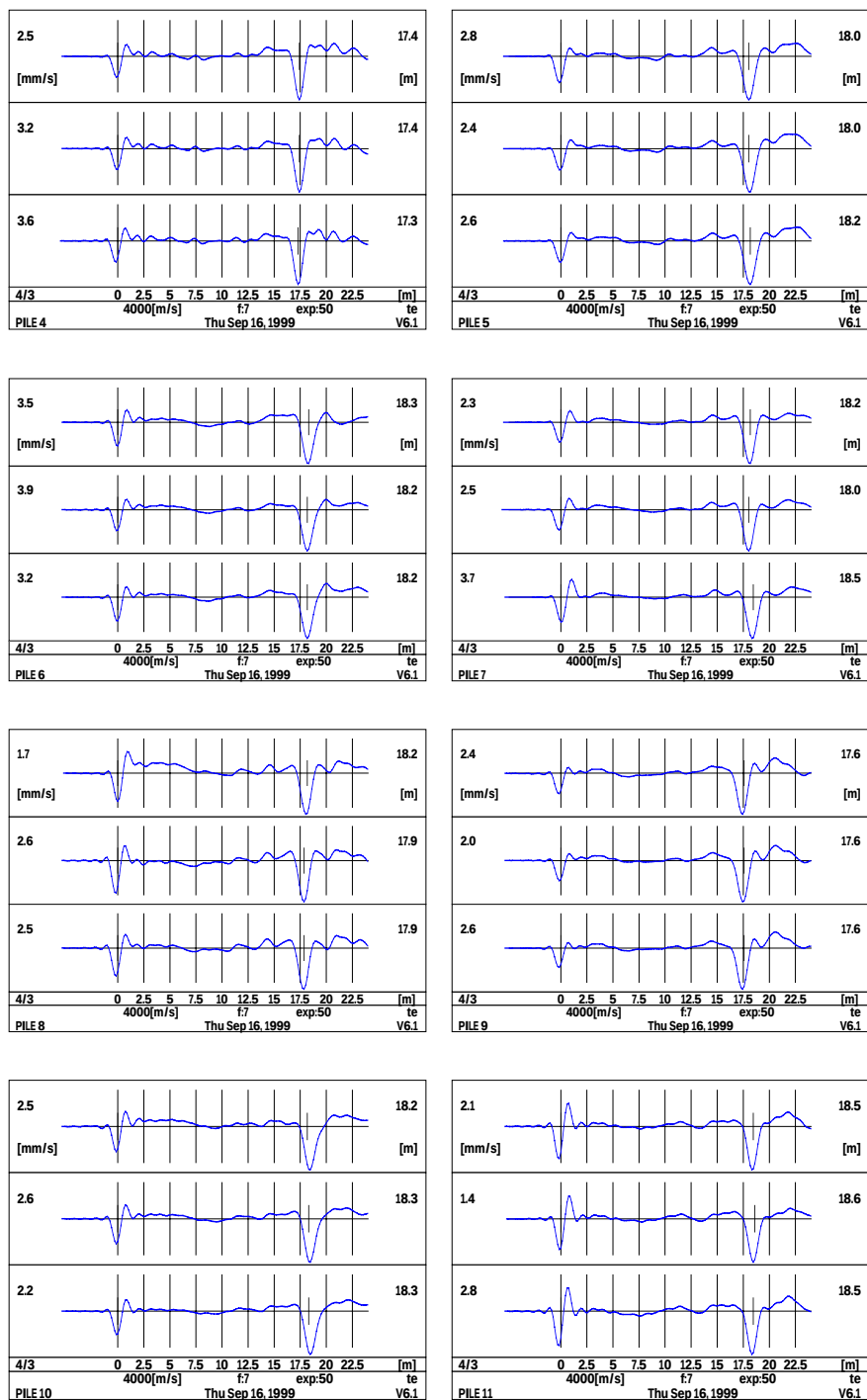


Figure 4.1 , Example of piles constructed by a similar installation method, in the same ground conditions, of similar age and having the same length.

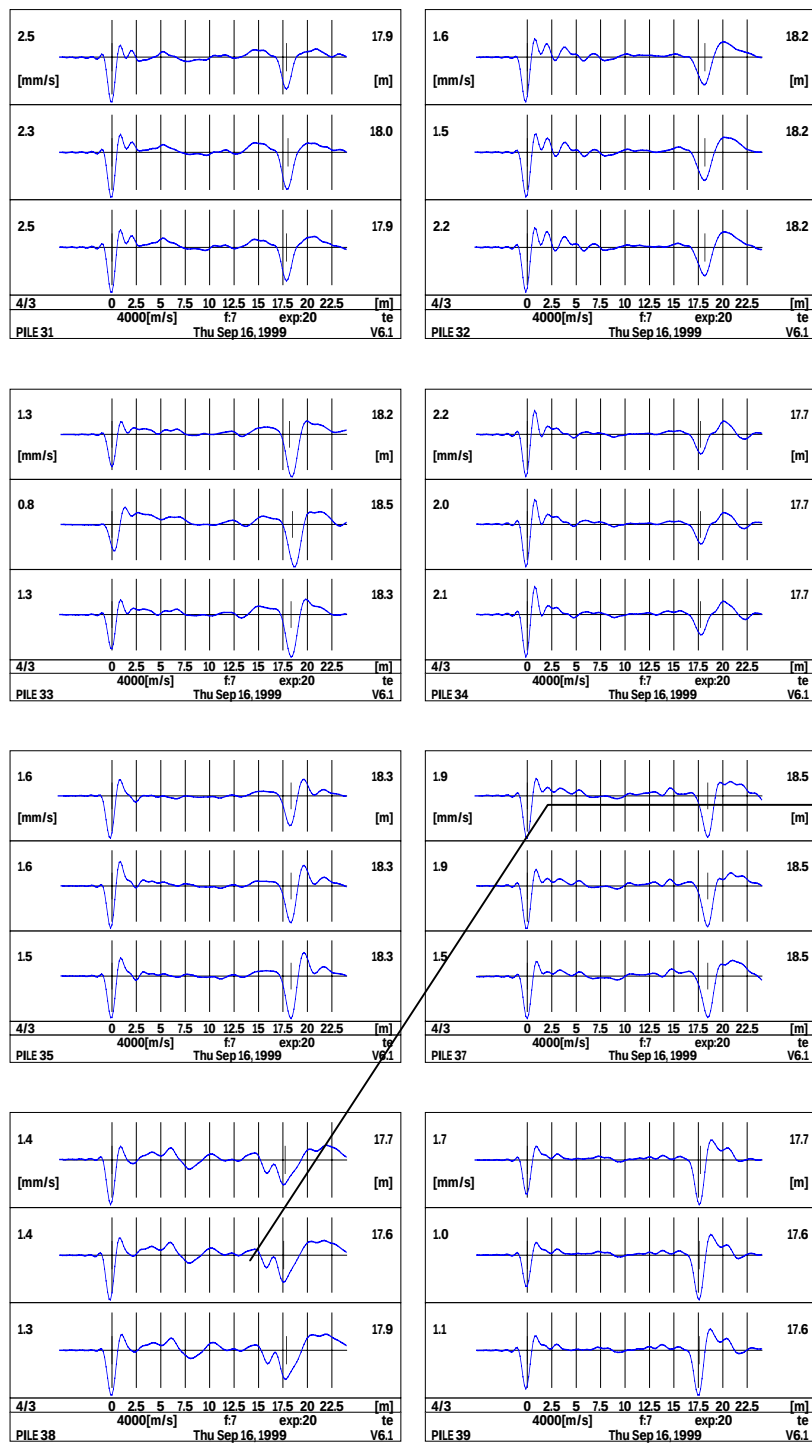


Figure 4.2, Example of pile with signal deviating from the characteristic signal

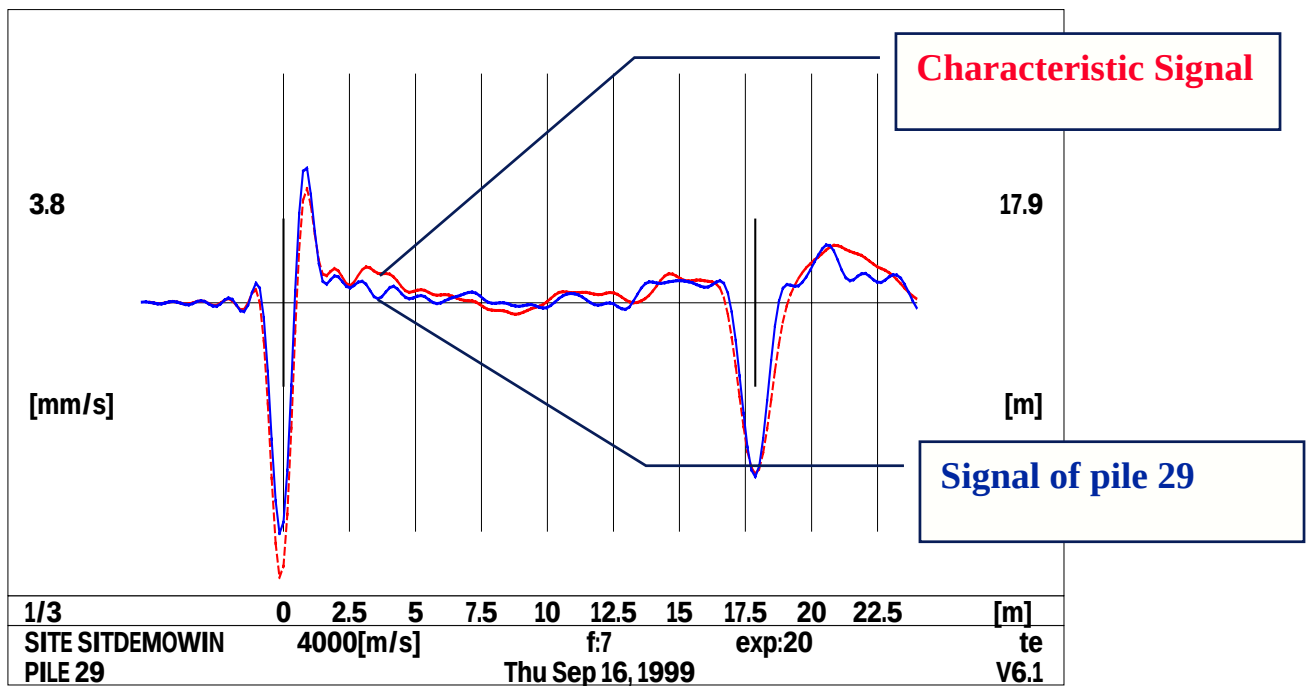


Figure 4.3 , Example of a SIT Signal corresponding with the characteristic signal

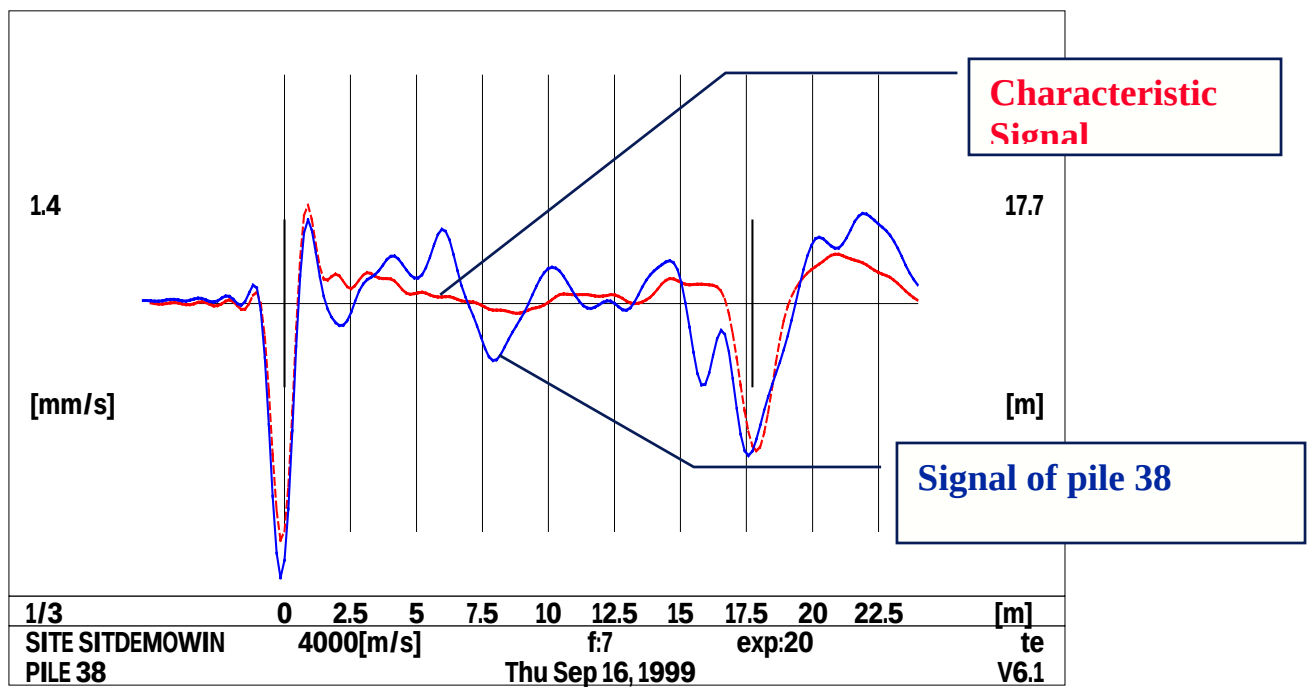


Figure 4.4, Example of a SIT Signal deviating from the characteristic signal



5 Interpretation Guidelines

- Test at least 20 % of the piles on site, with a minimum of 10 at several locations on a site.
- Never be satisfied testing only piles which appear suspect.
- Determine the characteristic signal for piles of the same type on a site. The **characteristic signal** from the site or group average should be compatible with that for other piles of the same type, and should generally correspond with the majority of piles tested on a particular site. The **characteristic signal** can be established intuitively or made by averaging a number of pile results together, excluding any piles which deviate from the norm.
- Compare the **characteristic signal** with the available soil data.
- Try to understand the causes of deviations—most often a change in pile cross section caused by soft layers, fill materials, voids in ground, old foundation bases, entry into hard layers, casing lengths, or deliberate pile base enlargements.

Note: If a pile enters a rock material, damping will be very high because of greatly increased shaft friction and this will show as an apparent increase in cross section and there will be no reaction from the pile toe.

- If possible, try to determine the pile length from the **characteristic signal**.
- Carry out individual pile interpretation using all individual signals.
- Flag as "suspect" signals with important deviations from normal.
- Determine the level and type of deviation from normal and physically examine the pile.

Note: Be aware that three dimensional effects will influence the signals for pile diameters larger than 0.4m.



6 Interpretation

For interpretation six classes are distinguished

Class 1 Pile OK

Class 2, Pile head problem

Class 3, Discontinuity reducing the nominal impedance of the pile shaft
(The nominal impedance represents normally the impedance at the pile top)

Class 4, Pile seems too short

Class 5, Pile seems too long

Class 6, Deviating or no pile toe reflection

Examples of results for each class are presented below

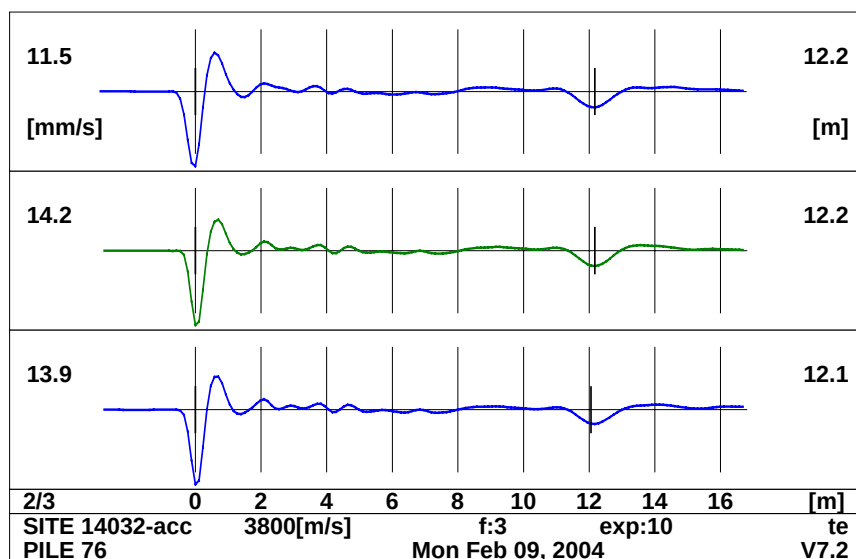


Figure 6.1. Class 1. Pile good, length OK,

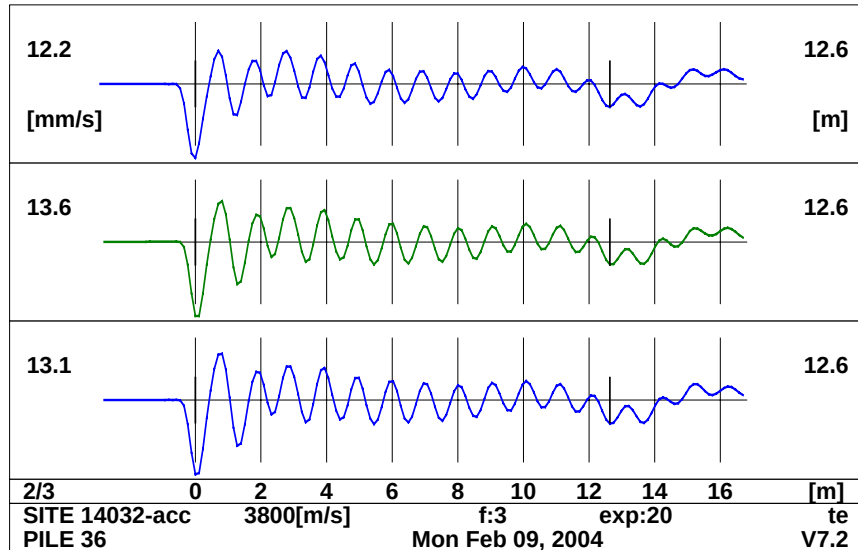


Figure 6.2 Pile head problem, Class 2
Possible reason, crack, necking, low material quality.

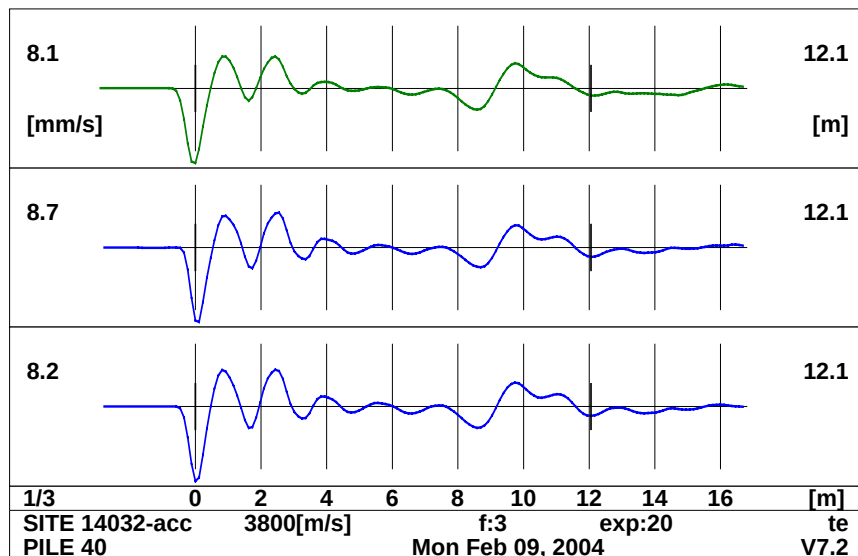


Figure 6.3. Class 3. Reflection from discontinuity at 1.5 m from the pile head.
Possible cause, crack, necking, low material quality.

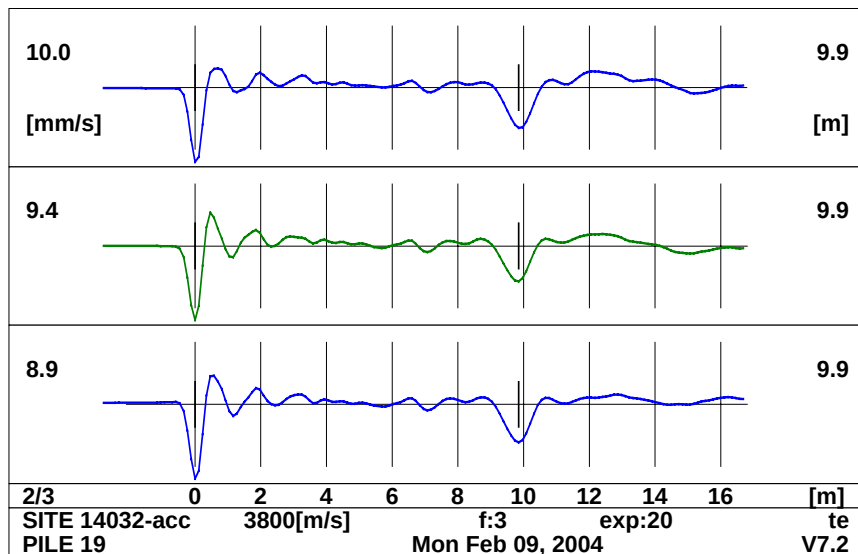


Figure 6.4. Class 4. Pile too short, reflection at 10m, should be 12m
Possible cause: made too short, broken, heavily cracked or serious necking at 10 from the top.

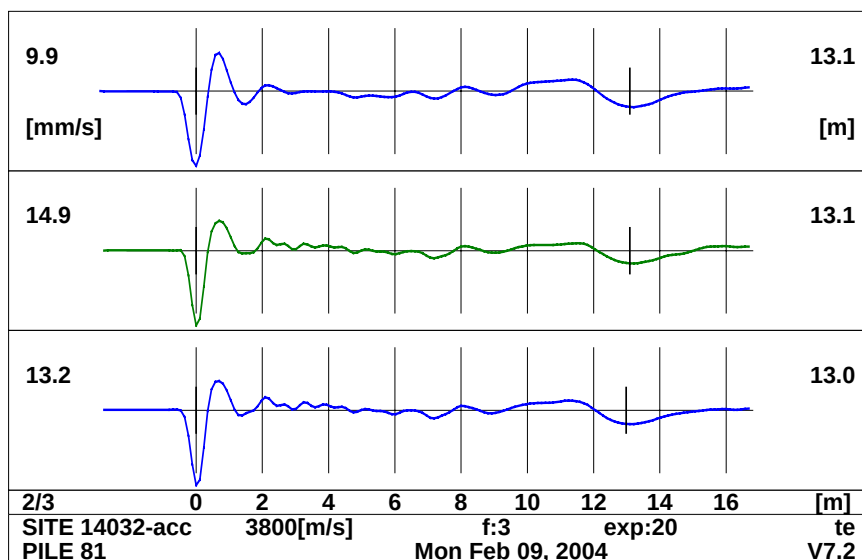


Figure 6.5. Class 5. Toe reflection too late.
Possible cause: Pile made too long, lower material quality, pile younger than other piles,

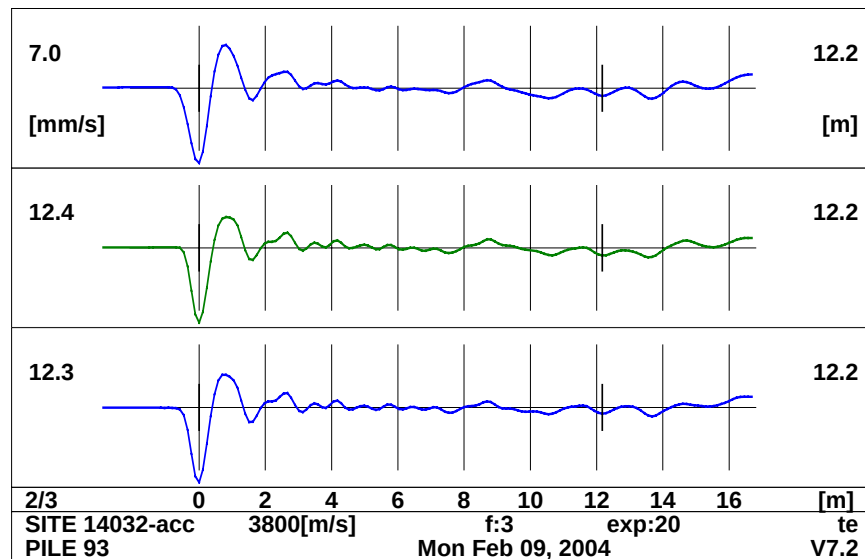


Figure 6.6. Class 6. No clear toe reflection
Possible cause, large friction, deviating material properties at pile toe

7 Pile preparation

Driven piles:

Cast-in-situ piles:

All piles:

Trim pile top if cracked

Remove soft pile top to sound concrete

Cut away overbreak or overspill at ground level.

Pile tops must be clean, though not necessarily flat and smooth



Figure 7.1 Example of pile top with sound concrete

8 Placing the Sensor

Place the accelerometer with a small amount of **Pietoplast** to the top of the pile, ensuring that the sensor is vertical. Choose a relatively flat, off-centre spot and press with hand weight. (The **Pietoplast** helps transfer the signal and damp the resonance from pile to sensor.) Do not position the sensor close to the location of impact.

Note: If the pile top is not easily accessible, the accelerometer can also be attached to the pile shaft.

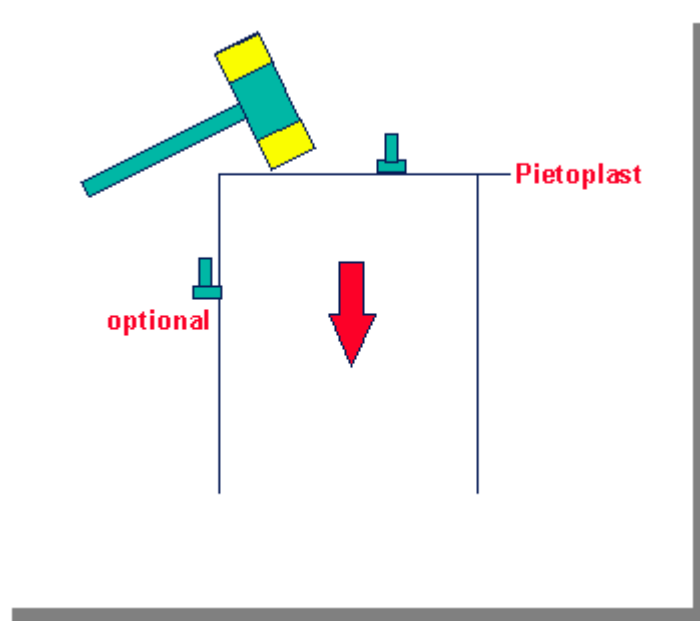


Figure 8.1 Example of placement of transducer

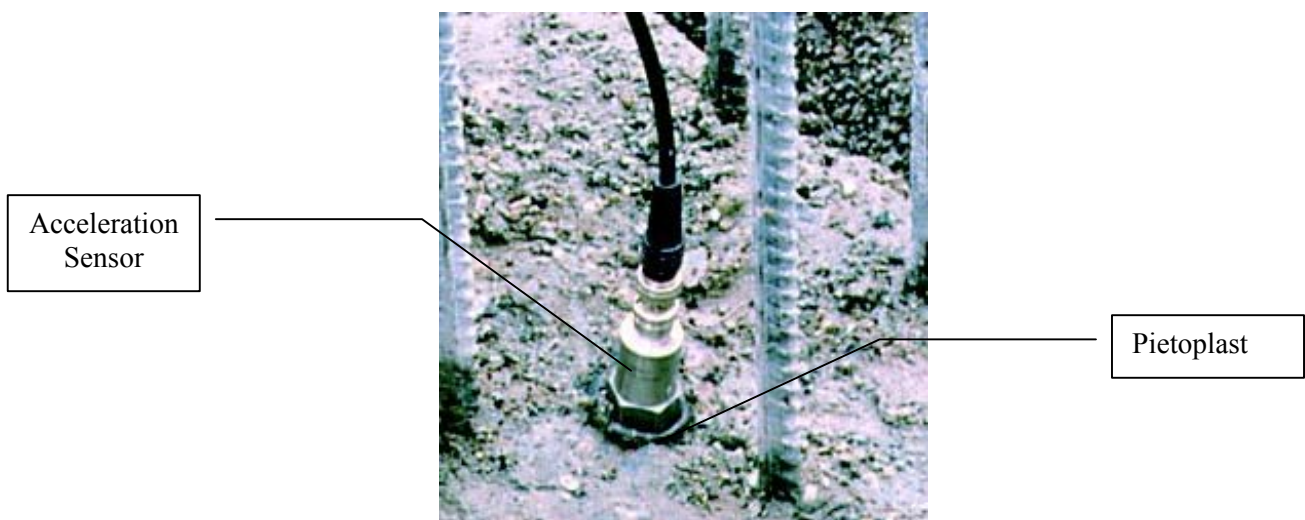


Figure 8.2 Example of placement of transducer



9 Reliable Testing

The requirements for reliable testing include:

- ☐ test **at least 20 %** of piles on site, with a minimum of 10
- ☐ for the highest **confidence**, test all piles
- ☐ use proper hammer with a **blow length** of 1 m or less (see Appendix).
- ☐ record at least **3 similar signals** per pile
- ☐ **process** all signals in a similar way, with **minimal filtering** and **optimal amplification**
- ☐ determine **characteristic signal**
- ☐ check **characteristic signal** with soil investigation data (see Appendix)
- ☐ check signals that deviate from the **characteristic signal**
- ☐ perform **qualitative** interpretation
- ☐ use all available information
- ☐ Follow the guidelines

If necessary, perform **quantitative** interpretation with integrity testing signal matching using **TNOWAVE** (SITWAVE). If there are still doubts, excavate the pile, do coring, or conduct a load test (DLT, STN, SLT) or reject the pile.

10 References

1. Ulrich, G., Stocker, M., (1983), Integritätsuntersuchung an präparierten Betonpfähle, (Integrity Testing Research on Prepared Concrete Piles), Symposium Messtechnik im Erd- und Grundbau, Germany
2. Reiding, F.J., Middendorp, P., P.J. van Brederode, 1984, A digital approach to sonic pile testing. 2nd International Conference on Stress Waves, The Hague, Netherlands, Balkema
3. Fleming, K, Reidng, F.J., Middendorp, P, 1985, Faults in cast in place piles and their detection. Proceedings of the Second International Conference on Structural Faults and Repair, The Institution of Civil Engineers, Westminster London.
4. Middendorp, P., Reiding, F.J., 1988, Determination of Discontinuities in Piles by TNO Integrity and Signal Matching Techniques, 3rd International Conference on The Application of Stress Wave Theory on Piles, Ottawa, Canada
5. Starke, W.F., Janes, M.C., 1988, Accuracy and Reliability of Low Strain Testing, 3rd International Conference on The Application of Stress Wave Theory on Piles, Ottawa, Canada
6. England, M., 1991, A guide to Low-Strain Integrity Testing , Report Cementation Piling and Foundations Ltd, Field Data Acquisition Department, United Kingdom.
7. ASTM D5882-96 (1996) Standard Test Method for Low Strain Integrity Testing of pile, American Society for Testing and Materials
8. Turner M.J., (1997) Integrity Testing In Piling Practice, CIRIA Report 144
9. German Society for Geotechniques (DGGT) (1998), Recommendations for Static and Dynamic Pile Tests ,
10. Chow, Y.K., Phoon, K.K., Chow, W.F., (2004) Three-Dimensional Stress Wave Analysis of Pile Integrity Tests, 7th International Conference on The Application of Stress Wave Theory on Piles, Kuala Lumpur, Malaysia
11. Seidel, J.P., Tan, S.K.,(2004), Elimination of the Rayleigh Wave Effect on Low Strain Integrity Test Results. 7th International Conference on The Application of Stress Wave Theory on Piles, Kuala Lumpur, Malaysia

11 Appendix, Influence of hammer blow width

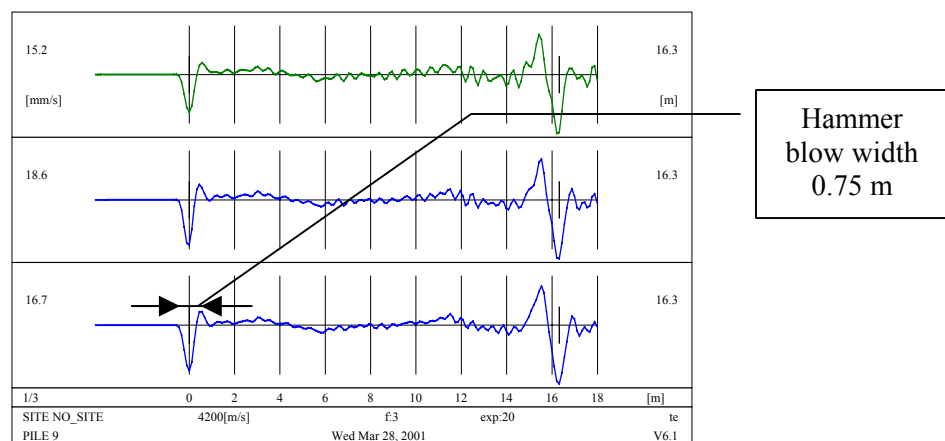
Reference:: Profound report 01-PROF-B0566, SIT User's day. 21 March 2001 (In Dutch)



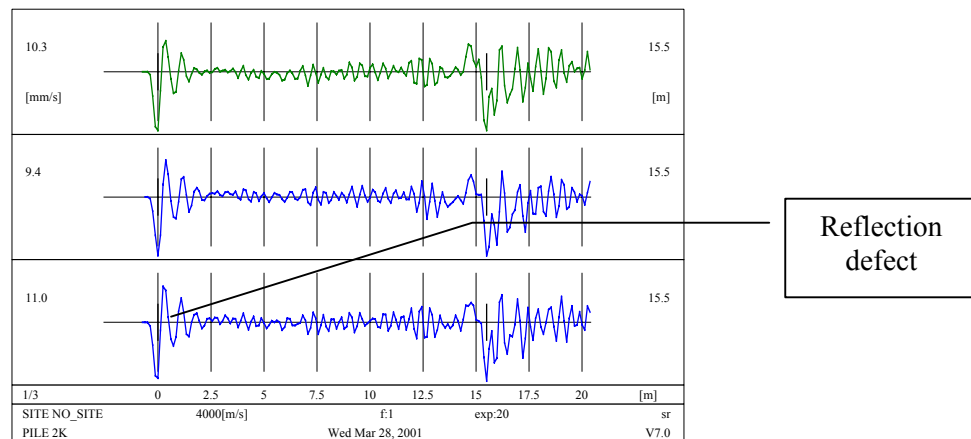
Pile 9, Reference pile, no defects



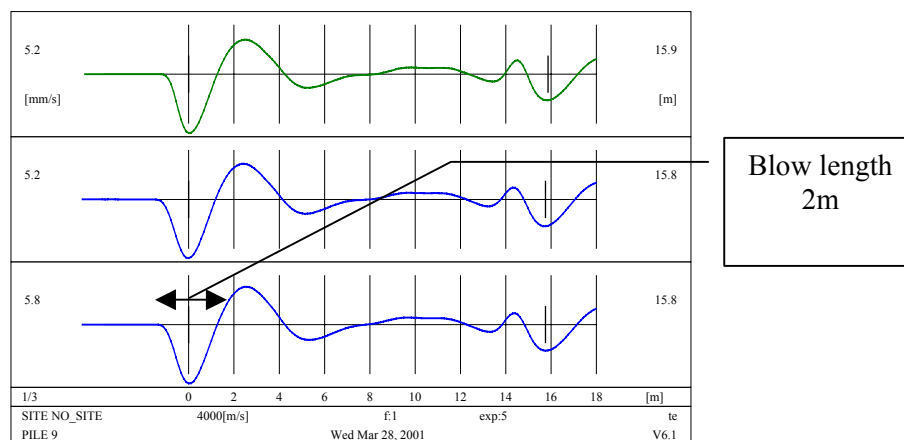
Pile 2, with 49% cross section reduction cut



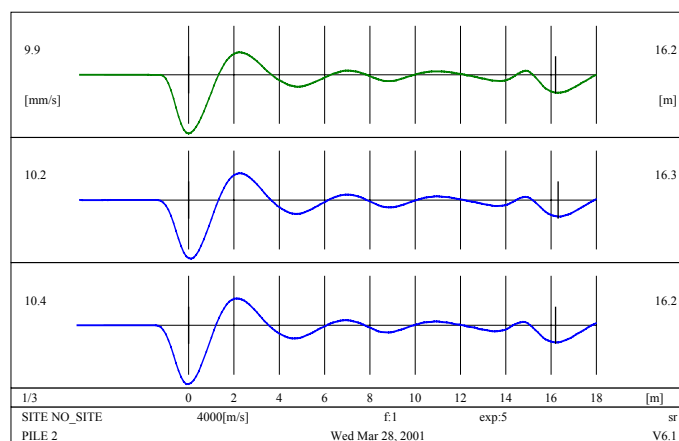
SIT Results Reference Pile 9, Hammer blow width 0.75m



SIT Result, Pile 2 , Hammer blow width 0.75m.
Defect visible



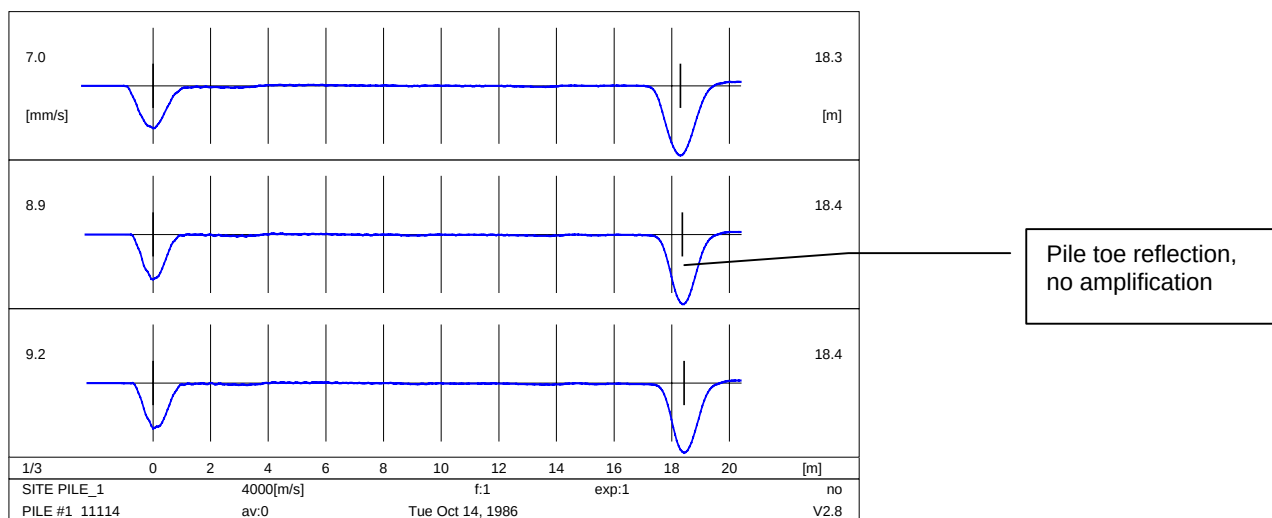
SIT Result, Reference Pile 9 , Hammer blow width 2 m,



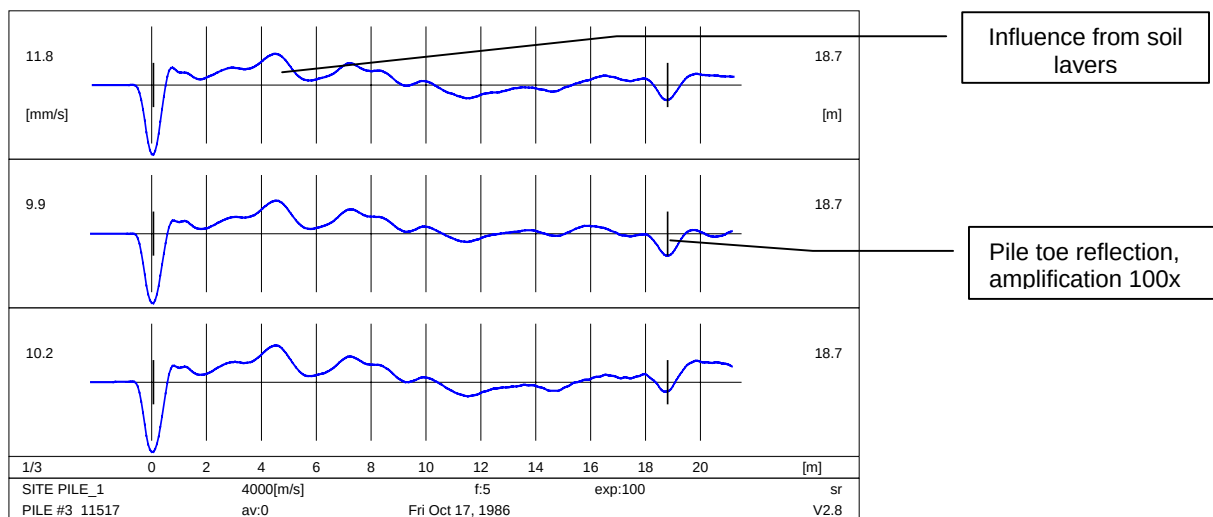
SIT Result, Pile 2 , Hammer blow width 2 m,
Defect not visible!!!!

12 Appendix, Influence of soil

The shaft friction reduces the amplitude of the stress wave. Stiff soil layers cause a higher damping than soft layers. This means that the detection of the pile toe reflection (pile length) has a limit. For very stiff clays only the first 5 m from the pile head yield information about the pile shaft. Pile toe reflections from 80 m have been detected in soft soils. In the Netherlands the pile toe reflection can be measured up to 35m.



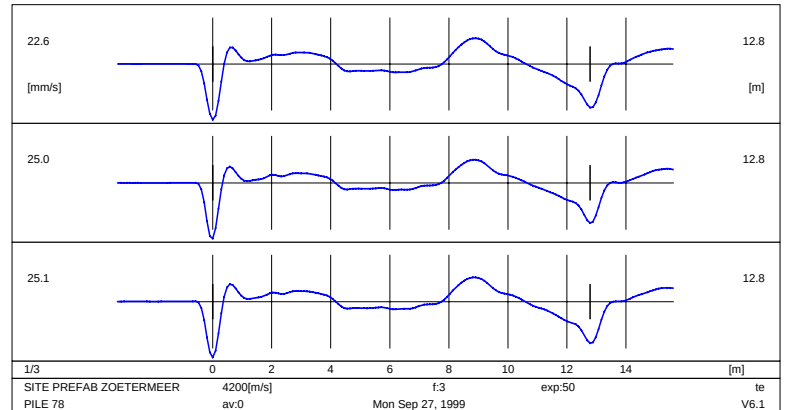
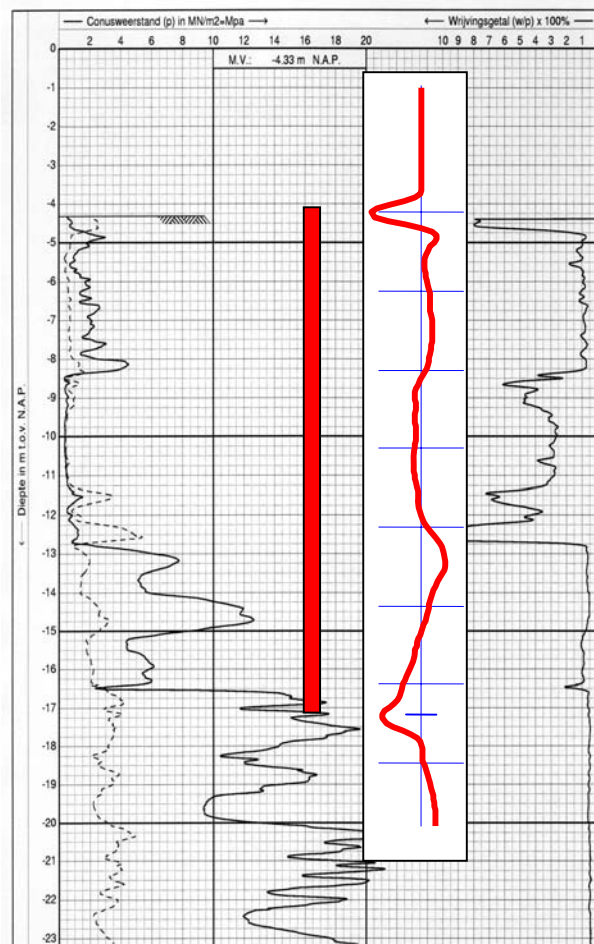
Precast pile tested before driving



Precast pile tested after driving

Soil layers changes act as discontinuities along the pile shaft and resulting reflections are visible in the signals. For this reason it is essential that soil investigation results are included in the interpretation.

Reflections from a greater depth are weaker and to make them visible the signals are amplified exponentially with time.



. Integrity testing results of cast in situ pile

. Integrity testing results and relation with soil investigation result. (CPT)